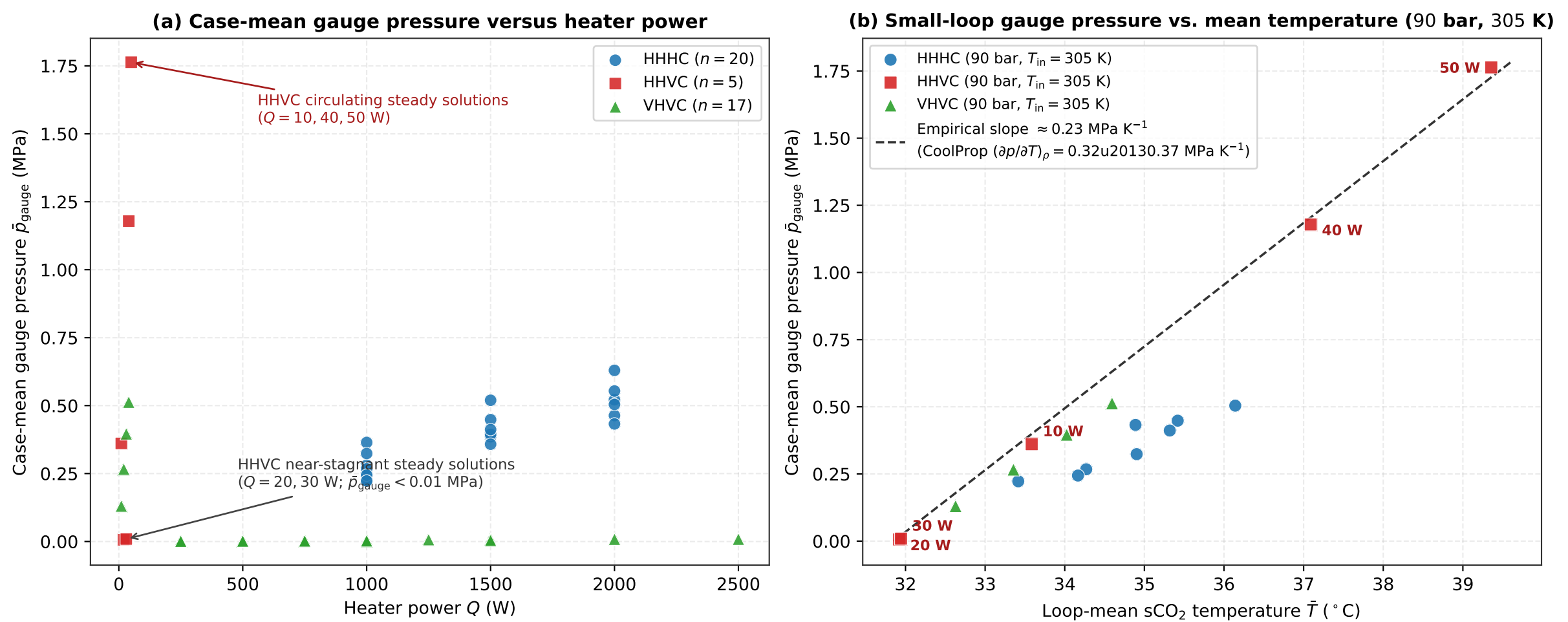
(a) Case-mean gauge pressure versus heater power
HHHC ($n = 20$)
HHVC ($n = 5$)
VHVC ($n = 17$)
HHVC circulating steady solutions
($Q = 10, 40, 50$ W)
HHVC near-stagnant steady solutions
($Q = 20, 30$ W; $\bar{p}_{gauge} < 0.01$ MPa)
Case-mean gauge pressure $\bar{p}_{gauge}$ (MPa)
Heater power Q (W)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0
500
1000
1500
2000
2500
(b) Small-loop gauge pressure vs. mean temperature (90 bar, 305 K)
HHHC (90 bar, $T_{in} = 305$ K)
HHVC (90 bar, $T_{in} = 305$ K)
VHVC (90 bar, $T_{in} = 305$ K)
Empirical slope ≈ 0.23 MPa K^{-1}
(CoolProp $(\partial p/\partial T)_\rho = 0.32u20130.37$ MPa K^{-1})
50 W
40 W
10 W
30 W
20 W
Case-mean gauge pressure $\bar{p}_{gauge}$ (MPa)
Loop-mean sCO$_2$ temperature $\bar{T}$ ($^\circ$C)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
32
33
34
35
36
37
38
39